1) Caesar cipher, key $k = 3$

Plaintext: **ATTACK AT DAWN**

Rule:

$$C = (P + 3) mod 26$$

Encryption:

- A → D
- T → W
- T → W
- A → D
- C → F
- K → N

So,

Ciphertext = DWWDFN DW GDZQ

2) Caesar ciphertext: UFTU NFTTBHF

Try all possible shifts.

Using Caesar decryption with all 26 keys:

- $k = 0$: UFTU NFTTBHF
- $k = 1$: **TEST MESSAGE**
- $k = 2$: SDRS LDRRZFD
- ...
- remaining shifts are meaningless compared to normal English text.

So the correct plaintext is:

TEST MESSAGE

Number of possible keys in Caesar cipher:

26 total shifts
If we exclude shift 0 as trivial, then **25 effective non-trivial keys**.

3) Caesar decryption of FYYFHP FY IFBS, key $k = 5$

Rule:

$$P = (C - 5) mod 26$$

Step-by-step:

- F → A
- Y → T
- Y → T
- F → A
- H → C
- P → K

So:

- FYYFHP → ATTACK
- FY → AT
- IFBS → DAWN

Plaintext = ATTACK AT DAWN

4) Multiplicative cipher, plaintext SAFE COMMUNICATION, key $k = 9$

Rule:
$C = (9P) mod 26$, with A=0, B=1, ..., Z=25

Letter conversions:

- S(18) → 9×18 = 162 mod 26 = 6 → G
- A(0) → 0 → A
- F(5) → 45 mod 26 = 19 → T

- E(4) → 36 mod 26 = 10 → K

So:

- SAFE → GATK

Similarly,

- COMMUNICATION → SWEEYNUSAPUWN

Ciphertext = GATK SWEEYNUSAPUWN

5) Multiplicative ciphertext UZZUYM UZ JUEH, key $\boldsymbol{k = 5}$

For decryption, inverse of 5 mod 26 is:

$5^{-1} = 21$, because $5 \times 21 = 105 \equiv 1(\mathrm{mod}26)$

Rule:

$$P = 21C mod 26$$

Decryption gives:

Decrypted text = EFFEKS EF HEGR

6) Affine cipher decryption of IHHWVCSWFRCP, with $\boldsymbol{a = 5, b = 8}$

Affine formulas:

- Encryption: $C = (aP + b) mod 26$
- Decryption: $P = a^{-1}(C - b) mod 26$

Here:

- $a = 5$
- $b = 8$
- inverse of 5 mod 26 = 21 (Find using extended Euclidian theorem)

So:

$$P = 21(C - 8) mod 26$$

Decryption result:

AFFINECIPHER

So the plaintext is: = **AFFINE CIPHER**

7) Affine encryption of SECURITY, with $a = 7, b = 3$

Rule:

$$C = (7P + 3) mod 26$$

Using A=0, ..., Z=25:

- S(18) → $7 \times 18 + 3 = 129 \text{mod } 26 = 25$→ Z
- E(4) → $28 + 3 = 31 \text{mod } 26 = 5$→ F
- C(2) → $14 + 3 = 17$→ R
- U(20) → $140 + 3 = 143 \text{mod } 26 = 13$→ N
- R(17) → $119 + 3 = 122 \text{mod } 26 = 18$→ S
- I(8) → $56 + 3 = 59 \text{mod } 26 = 7$→ H
- T(19) → $133 + 3 = 136 \text{mod } 26 = 6$→ G
- Y(24) → $168 + 3 = 171 \text{mod } 26 = 15$→ P

Ciphertext = ZFRNSHGP

8) Playfair cipher

Keyword: CYBERSECURITY
Plaintext: DEFEND THE NETWORK

Using I/J combined, the 5×5 matrix becomes:

C	**Y**	**B**	**E**	**R**
S	U	I	T	A
D	F	G	H	K
L	M	N	O	P
Q	V	W	X	Z

Prepared plaintext digraphs:
DE FE ND TH EN ET WO RK

Encrypting pairwise:

- DE → HC
- FE → HY
- ND → LG
- TH → HO
- EN → BO
- ET → TH
- WO → XN
- RK → AP

Ciphertext = HC HY LG HO BO TH XN AP

9) Alphanumeric Playfair-style cipher

Keyword: SECURE2026
Plaintext: ATTACK AT 2100 HOURS

Using a 6×6 matrix with A–Z and 0–9:

S	**E**	**C**	**U**	**R**	**2**
0	6	A	B	D	F
G	H	I	J	K	L
M	N	O	P	Q	T
V	W	X	Y	Z	1
3	4	5	7	8	9

Prepared digraphs:
AT TA CK AT 21 0X 0H OU RS

Encryption:

- AT → FO
- TA → OF
- CK → RI
- AT → FO
- 21 → F9
- 0X → AV
- 0H → 6G
- OU → PC
- RS → 2E

Ciphertext = FO OF RI FO F9 AV 6G PC 2E

Q10) Decrypt R8Y$ ZK1@ QP7L X3#F

Keyword: CYBERSECURITY2026@#
Symbols used: @ # $ % & * + - = ? ! _ .

Key Matrix (7 × 7)

C	**Y**	**B**	**E**	**R**	**S**	**U**
I	T	2	0	6	@	#
A	D	F	G	H	J	K
L	M	N	O	P	Q	V
W	X	Z	1	3	4	5
7	8	9	$	%	&	*
+	-	=	?	!	_	.

Ciphertext Pairs

R8 | Y$ | ZK | 1@ | QP | 7L | X3 | #F

Using **Playfair-style rules**:

- Same row → move **left**
- Same column → move **up**
- Rectangle → swap columns

Decryption

R8 → Y%
Y$ → E8
ZK → 5F
1@ → 40
QP → PO
7L → WA
X3 → W1
#F → 2K

Final Plaintext = Y%E85F40POWAW12K

11) Hill Cipher Encryption for SAFE

Key matrix

$$K = \begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix}$$

Alphabet mapping

$$A = 0, B = 1, \ldots, Z = 25$$

Hill Cipher encryption equation:

$$C = KP(\text{mod}26)$$

Plaintext blocks

SAFE → divide into pairs

SA , FE

$$SA = \begin{bmatrix} 18 \\ 0 \end{bmatrix}$$
$$FE = \begin{bmatrix} 5 \\ 4 \end{bmatrix}$$

Block 1

$$C = KP(\text{mod}26)$$
$$\begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} 18 \\ 0 \end{bmatrix} = \begin{bmatrix} 198 \\ 54 \end{bmatrix}$$
$$\begin{bmatrix} 198 \\ 54 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 16 \\ 2 \end{bmatrix}$$

16 → Q
2 → C

Result = **QC**

Block 2

$$\begin{bmatrix} 11 & 8 \\ 3 & 7 \end{bmatrix} \begin{bmatrix} 5 \\ 4 \end{bmatrix} = \begin{bmatrix} 87 \\ 43 \end{bmatrix}$$
$$\begin{bmatrix} 87 \\ 43 \end{bmatrix} \text{mod } 26 = \begin{bmatrix} 9 \\ 17 \end{bmatrix}$$

9 → J
17 → R

Result = **JR**

Final Ciphertext = QCJR

12) 3×3 Hill Cipher for ACTNOW

Key matrix

$$K = \begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix}$$

Encryption equation

$$C = KP(\text{mod}26)$$

Plaintext blocks

ACT , NOW

$$ACT = \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix}$$
$$NOW = \begin{bmatrix} 13 \\ 14 \\ 22 \end{bmatrix}$$

Block 1

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 0 \\ 2 \\ 19 \end{bmatrix} = \begin{bmatrix} 67 \\ 222 \\ 319 \end{bmatrix}$$
$$\text{mod } 26 = \begin{bmatrix} 15 \\ 14 \\ 7 \end{bmatrix}$$

15 → P
14 → O
7 → H

Result = **POH**

Block 2

$$\begin{bmatrix} 6 & 24 & 1 \\ 13 & 16 & 10 \\ 20 & 17 & 15 \end{bmatrix} \begin{bmatrix} 13 \\ 14 \\ 22 \end{bmatrix} = \begin{bmatrix} 435 \\ 611 \\ 830 \end{bmatrix}$$
$$\text{mod } 26 = \begin{bmatrix} 20 \\ 15 \\ 22 \end{bmatrix}$$

20 → U
15 → P
22 → W

Result = **UPW**

Final Ciphertext = POHUPW

13) 4×4 Hill Cipher for DEFENDTHEBASE

Key matrix

$$K = \begin{bmatrix} 3 & 10 & 20 & 5 \\ 20 & 9 & 17 & 8 \\ 9 & 4 & 17 & 6 \\ 15 & 14 & 21 & 1 \end{bmatrix}$$

Encryption equation

$$C = KP(\text{mod}26)$$

Plaintext

DEFENDTHEBASE

Since block size = 4

Pad with X

DEFENDTHEBASEXXX

Blocks

DEFE
NDTH
EBAS
EXXX

Apply

$$C = KP \bmod 26$$

to each block.

Final ciphertext = **NFWCQQATIZSOLEHE**

14) Polyalphabetic Cipher

Plaintext

NETWORK SECURITY

Keyword

SAFE

Encryption equation

$$C_i = (P_i + K_i) \bmod 26$$

Key repetition

SAFE SAFE SAFE SA

Example calculation

$$N(13) + S(18) = 31$$
$$31 \bmod 26 = 5 = F$$

Continue for all characters.

Final ciphertext = **FEYAGRP WWCZVATD**

15) Vigenère Cipher

Plaintext

INFORMATION SECURITY

Keyword

NETWORK

Encryption equation

$$C_i = (P_i + K_i) \bmod 26$$

Decryption equation

$$P_i = (C_i - K_i) \bmod 26$$

Key repetition

NETWORKNETWORKNETW...

Example

$$I(8) + N(13) = 21$$
$$21 \bmod 26 = V$$

Final ciphertext = **VRYKFDKGMHJ GVMHVBPM**

16) Columnar Transposition Cipher

Keyword: CIPHER
Plaintext: DEFEND THE NETWORK

Remove spaces:

Plaintext = DEFENDTHENETWORK

Step 1: Number the key letters in alphabetical order

Keyword letters: C I P H E R

Alphabetical order:

- **C → 1**
- **E → 2**
- **H → 3**
- **I → 4**

- **P → 5**
- **R → 6**

So the numbering becomes:

C I P H E R

1 4 5 3 2 6

Step 2: Write plaintext row-wise under the key

C	**I**	**P**	**H**	**E**	**R**
D	E	F	E	N	D
T	H	E	N	E	T
W	O	R	K		

Step 3: Read columns in the alphabetical order of the key

Column order from numbering:

$$\mathbf{1, 2, 3, 4, 5, 6}$$

That means read columns as:

C, E, H, I, P, R

Now read each column top to bottom:

- **C → DTW**
- **E → NE**
- **H → ENK**
- **I → EHO**
- **P → FER**
- **R → DT**

Step 4: Combine all columns

$$\text{Ciphertext} = \text{DTW} + \text{NE} + \text{ENK} + \text{EHO} + \text{FER} + \text{DT}$$

$$\boxed{\text{Ciphertext} = \text{DTWNEENKEHOFERDT}}$$

Final Answer = $\boxed{\text{DTWNEENKEHOFERDT}}$

17) Columnar Transposition Decryption

Updated Ciphertext: TKAATNTAWACD
Keyword: SAFE

Step 1: Find number of rows

Ciphertext length = 12
Key length = 4

$$\text{Rows} = \frac{12}{4} = 3$$

So we form a 3 × 4 matrix.

Step 2: Determine column order from keyword

Keyword = SAFE

Alphabetical order of letters:

A, E, F, S

So numbering is:

- **A → 1**
- **E → 2**
- **F → 3**
- **S → 4**

Hence,

S A F E

4 1 3 2

Step 3: Divide ciphertext into column groups

Since each column has 3 letters:

TKAATNTAWACD → TKA ATN TAW ACD

Now place these groups according to alphabetical order of the key:

- **A column = TKA**
- **E column = ATN**
- **F column = TAW**
- **S column = ACD**

So the table becomes:

S	**A**	**F**	**E**
A	T	T	A
C	K	A	T
D	A	W	N

Step 4: Read row-wise

Row 1 → ATTA
Row 2 → CKAT
Row 3 → DAWN

Therefore,

Final Answer = ATTACKATDAWN